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Data Sheet

Three-phase 250V gate driver

FD6288

Shenzhen Fengqi Technology Co., Ltd.

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Table of contents

1 System Introduction..... 3
1.1 Overview..... 3
1.2 Packaging..... 3
1.3 Product Features..... 3
1.4 Application scenarios..... 3
1.5 Typical Application Circuit..... 4
1.6 Functional Block Diagram..... 5
1.7 Pin Definition..... 6
1.7.1 FD6288T TSSOP20 Pin Diagram..... 6
1.7.2 FD6288T TSSOP20 Pin List..... 6
1.7.3 FD6288Q QFN24 Pin Diagram..... 7
1.7.4 FD6288Q QFN24 Pin List..... 7
2 Packaging Information..... 8
2.1 FD6288T TSSOP20..... 8
2.2 FD6288Q QFN24 9
3 Electrical Characteristics.....10
3.1 Absolute Maximum Ratings.....10
3.2 Recommended Operating Conditions.....10
3.3 Static electrical parameters.....11
3.4 Dynamic Electrical Parameters.....12
4 Logic Function Timing Diagram.....13
5 Transmission Time Test Standards.....13
6 Transmission Time Matching Test Standards.....13
7 Through-prevention function.....14
8 Deadband Function.....14
9 Modification Records.....22

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FD6288 three-phase 250V gate driver

1 System Introduction

1.1 Overview

FD6288 is a device that integrates three independent half-bridge gate Extreme drive integrated circuit chip, designed for high voltage and high speed drive MOSFET design, can operate up to +250V.

FD6288 has built-in VCC/VBS undervoltage (UVLO) protection function It can prevent the power tube from working at too low a voltage.

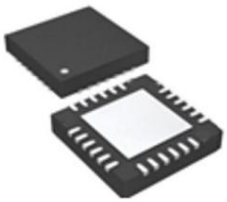
FD6288 has built-in shoot-through prevention and dead time to prevent The high and low side MOSFETs of the driver are directly connected to effectively protect the power device Piece.

FD6288 has built-in input signal filtering to prevent input noise interference.

1.2 Packaging



TSSOP20



QFN24

1.3 Product Features

ÿ Suspension absolute voltage +250V

ÿ Supply voltage operating range: 5.0 ÿ 20V

ÿ Integrates three independent half-bridge drivers

ÿ Output current +1.5A/-1.8A

ÿ 3.3V/5V input logic compatible

ÿ VCC/VBS undervoltage protection (UVLO)

ÿ Built-in shoot-through prevention function

ÿ Built-in 200ns dead time

ÿ Built-in input filtering function

ÿ High and low end channel matching

ÿ Output is in phase with input

1.4 Application Scenarios

Motor drive

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1.5 Typical Application Circuit

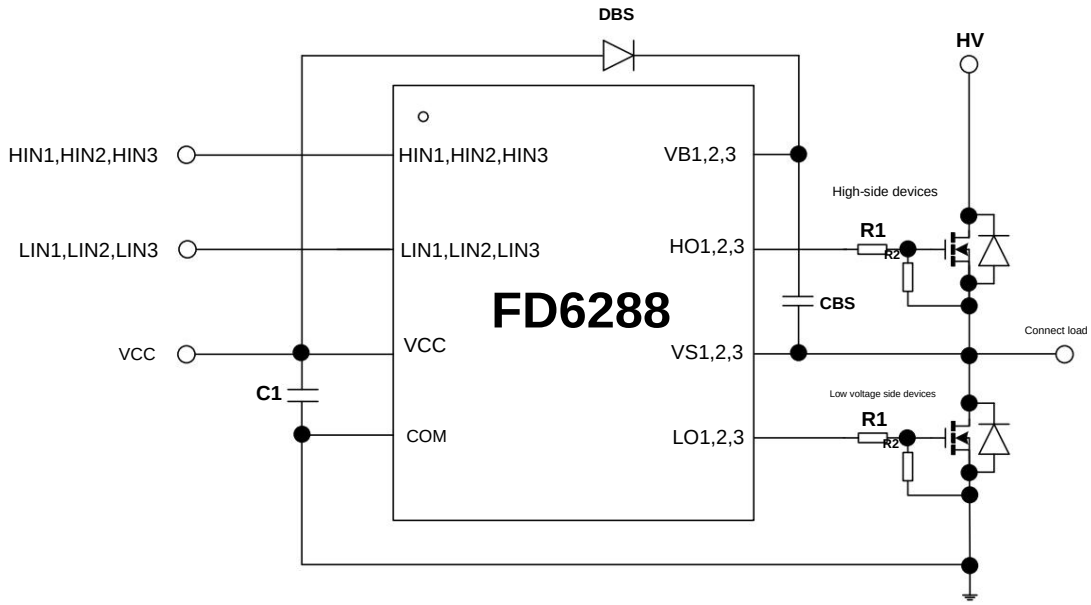


Figure 1-1 FD6288 typical application circuit

C1: Power filter capacitor, 10ÿF can be selected, as close to the chip pin as possible.

R1: Gate drive resistor, the resistance value depends on the driven device and dead time.

R2: MOS gate and source resistance.

Dbs: Bootstrap diode, a Schottky diode with high reverse breakdown voltage and shortest possible recovery time should be selected.

Cbs: Bootstrap capacitor, ceramic capacitor or tantalum capacitor should be selected and placed as close to the chip pin as possible.

Note: The above circuits and parameters are for reference only. The actual application circuit parameters should be set according to the measured results.

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1.6 Functional Block Diagram

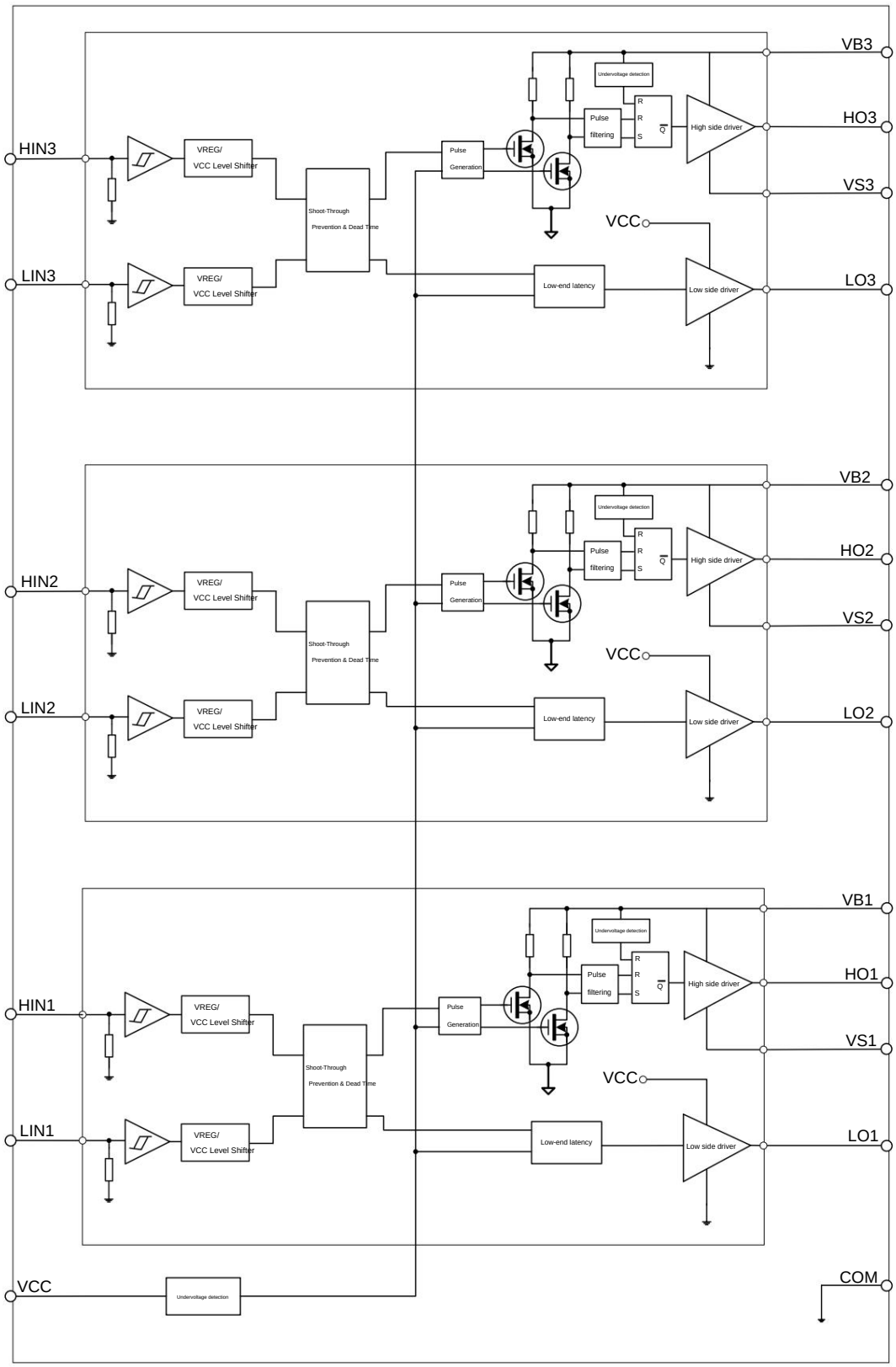


Figure 1-2 FD6288 functional block diagram

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1.7 Pin Definition

1.7.1 FD6288T TSSOP20 Pinout

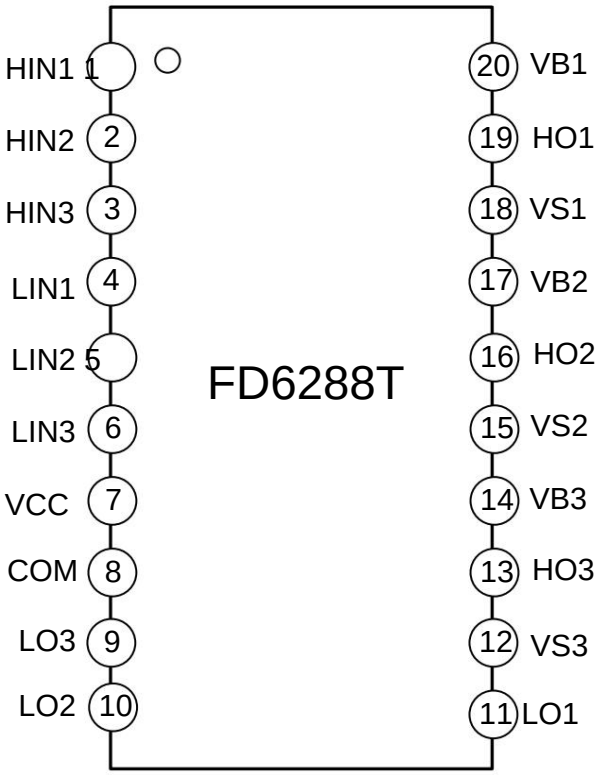


Figure 1-3 FD6288T TSSOP20 pin diagram

1.7.2 FD6288T TSSOP20 Pin List

Table 1-1 FD6288T TSSOP20 pin list

Pin Number	Pin Name	Pin Description
1, 2, 3	HIN1, HIN2, HIN3	High-side input
4, 5, 6	LIN1, LIN2, LIN3	low-side input
7	VCC	Low side supply voltage
8	COM	Grounding
9, 10, 11	LO3, LO2, LO1	low-side output
12, 15, 18	VS3, VS2, VS1	High-side floating offset voltage
13, 16, 19	HO3, HO2, HO1	High-side output
14, 17, 20	VB3, VB2, VB1	High side floating absolute voltage

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1.7.3 FD6288Q QFN24 Pinout

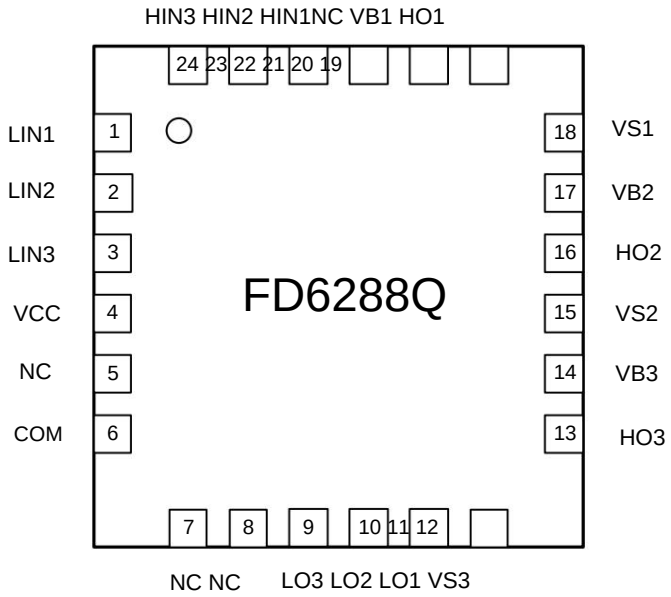


Figure 1-4 FD6288Q QFN24 pin diagram

1.7.4 FD6288Q QFN24 Pin List

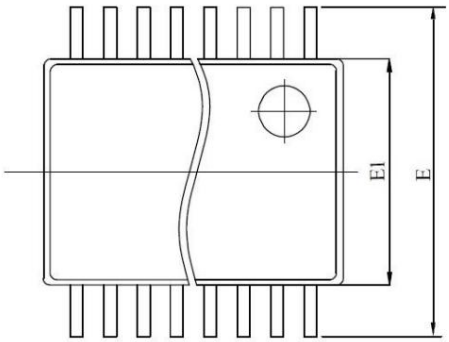
Table 1-2 FD6288Q QFN24 pin list

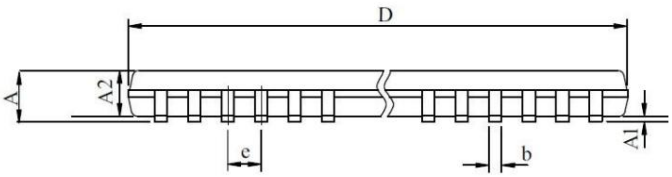
Pin Number	Pin Name	Pin Description
22, 23, 24	HIN1, HIN2, HIN3	High-side input
1, 2, 3	LIN1, LIN2, LIN3	low-side input
4	VCC	Low side supply voltage
6	COM	Grounding
9, 10, 11	LO3, LO2, LO1	low-side output
12, 15, 18	VS3, VS2, VS1	High-side floating offset voltage
13, 16, 19	HO3, HO2, HO1	high side output
14, 17, 20	VB3, VB2, VB1	High side floating absolute voltage
5, 7, 8, 21	NC	NC

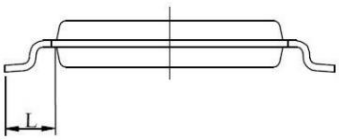
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2 Packaging Information

2.1 FD6288T TSSOP20





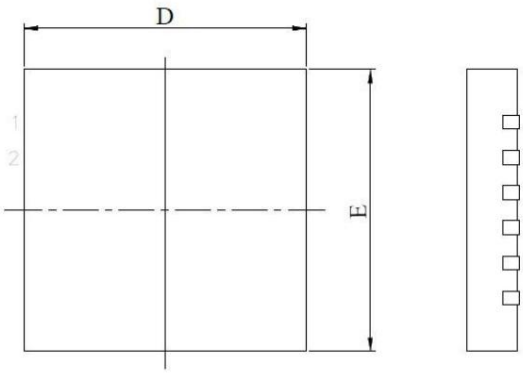


TSSOP20			
SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A			1.2
A1	0.05		0.15
A2	0.8	1	1.05
E	6.2	6.4	6.6
E1	4.3	4.4	4.5
D	6.4	6.5	6.6
L	0.45	0.6	0.75
e		0.65	
b	0.2		0.28

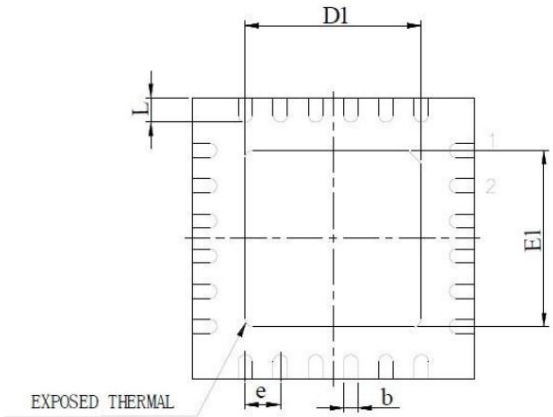
Product Name	Package	Order Model	Packaging	quantity
FD6288T	TSSOP20	FD6288T	Tape & Reel	3000

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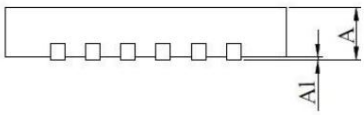
2.2 FD6288Q QFN24



TOP VIEW



BOTTOM VIEW



SIDE VIEW

QFN24L			
SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.7	0.75	0.8
A1	0		0.05
E	3.9	4	4.1
D	3.9	4	4.1
E1	2.55		2.8
D1	2.55		2.8
L	0.3	0.4	0.5
e		0.5	
b	0.2		0.3

Product Name	Package	Order Model	Packaging	quantity
FD6288Q	QFN24	FD6288Q	Tape & Reel	3000

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3 Electrical characteristics

3.1 Absolute Maximum Ratings

Table 3-1 Absolute maximum ratings

(Unless otherwise specified, all pins use COM as the reference point)

parameter		symbol	scope	unit
High side floating absolute voltage		VB1,2,3	-0.3ÿ275	V
High side floating offset voltage		VS1,2,3	VB1,2,3-25ÿVB1,2,3+0.3	V
High side output voltage		VHO1,2,3	VS1,2,3-0.3ÿVB1,2,3+0.3	V
Low side supply voltage		VCC	-0.3ÿ25	V
Low side output voltage		VLO1,2,3	-0.3ÿVCC+0.3	V
Logic input voltage (HIN, LIN)		VIN	-0.3ÿVCC+0.3	V
Offset voltage slew rate range		dVS/dt	ÿ50	V/ns
Power dissipation @TAÿ25ÿC	TSSOP20	PD	ÿ1.25	W
	QFN24		ÿ3.0	W
Thermal resistance from junction to ambient	TSSOP20	JbXj	ÿ100	ÿC/W
	QFN24		ÿ42	ÿC/W
Junction temperature range		Tj	ÿ150	ÿC
Storage temperature range		Tsj	-55ÿ150	ÿC

Note:

ÿ Do not exceed PD under any circumstances .

ÿ Voltage exceeding the absolute maximum rating may damage the chip.

3.2 Recommended operating conditions

Table 3-2 Recommended operating conditions[1]

(All voltages are referenced to COM)

parameter	symbol	Minimum	Maximum	unit
High side floating absolute voltage	VB1,2,3	VS1,2,3 + 5.0	VS1,2,3 + 20	V
High side floating offset voltage[2]	VS1,2,3	COM-4 [3]	250	V
High side output voltage	VHO1,2,3	VS1,2,3	VB1,2,3	V
Low side supply voltage	VCC	5.0	20	V
Low side output voltage	VLO1,2,3	0	VCC	V
Logic input voltage (HIN, LIN)	VIN	0	VCC	V
Ambient temperature	TA	-40	125	ÿC

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Note:

[1] Operating the chip above the recommended operating conditions for a long time may affect its reliability. It is not recommended to operate the chip outside the recommended operating conditions.

On long-term work.

[2] VS high-side floating offset voltage is tested at 15V bias voltage.

[3] When VS1,2,3 is (COM-4V) to 250V, HO works normally.

3.3 Static electrical parameters

Table 3-3 Static electrical parameters

(Unless otherwise specified, TA = 25°C, VCC = VBS1, 2, 3 = 15V, VS = COM)

parameter	symbol	Test conditions	Min	Typ	Max	Unit
High level input threshold voltage	VIH		2.7	-	-	V
Low level input threshold voltage	VIL		-	-	0.8	V
VCC undervoltage protection trip voltage	VCCUV+		4.2	4.6	5.0	V
VCC undervoltage protection reset voltage	VCCUV-		3.9	4.3	4.7	V
VCC undervoltage protection hysteresis voltage	VCC		0.2	0.3	-	V
VBS undervoltage protection trip voltage	VBSUV+		4.2	4.6	5.0	V
VBS undervoltage protection reset voltage	VBSUV-		3.9	4.3	4.7	V
VBS undervoltage protection hysteresis voltage	VBSUVH		0.2	0.3	-	V
Floating power supply leakage current	ILK	VB1,2,3=VS1,2,3=250V	-	0.1	5.0	ÿA
VBS Quiescent Current	IQBS	VIN=0V or 5V	-	180	270	ÿA
VBS dynamic current	IPBS	fHIN1,2,3=20kHz	-	180	270	ÿA
VCC Quiescent Current	IQCC	VIN = 0V or 5V	-	330	500	ÿA
VCC dynamic current	IPCC	fLIN1,2,3=20kHz	-	330	500	ÿA
LIN high level input bias current	ILIN+	VLIN=5V	-	25	40	ÿA
LIN low level input bias current	ILIN-	VLIN=0V	-	-	1	ÿA
HIN high level input bias current	IHIN+	VHIN=5V	-	25	40	ÿA
HIN low level input bias current	IHIN-	VHIN=0V	-	-	1	ÿA
Input pull-down resistor	RIN		140	200	260	Kÿ
High level output voltage	VOH	IO=100mA	-	0.6	0.9	V
Low level output voltage	VOL	IO=100mA	-	0.3	0.45	V
High level output short circuit pulse current	IOH	VO=0V, VIN=5V, PWDÿ10ÿs	1.1	1.5	1.9	A
Low level output short circuit pulse current	IOL	VO=15V, VIN=0V, PWDÿ10ÿs	1.3	1.8	2.3	A
VS static negative pressure	VSN		-	-6.0	-	V

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3.4 Dynamic electrical parameters

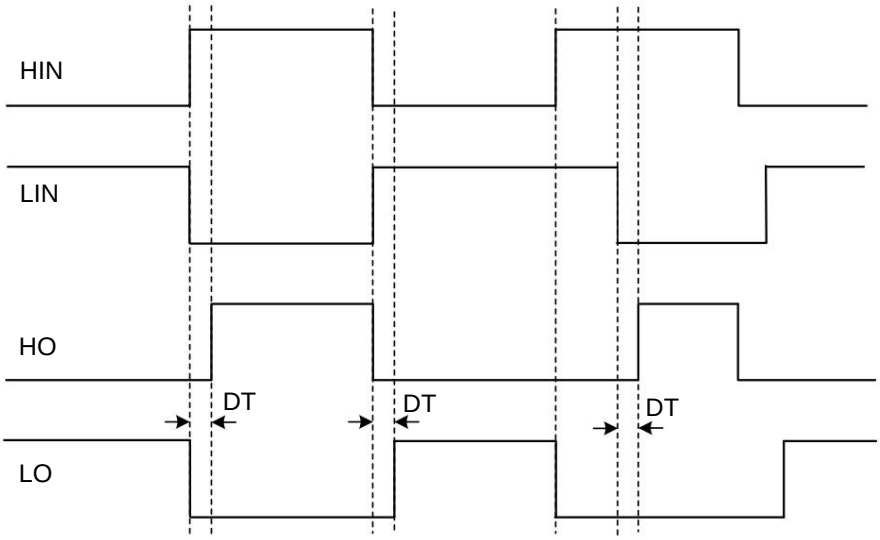
Table 3-4 Dynamic electrical parameters

(Unless otherwise specified, TA = 25°C, VCC = VBS1, 2, 3 = 15V, VS = COM)

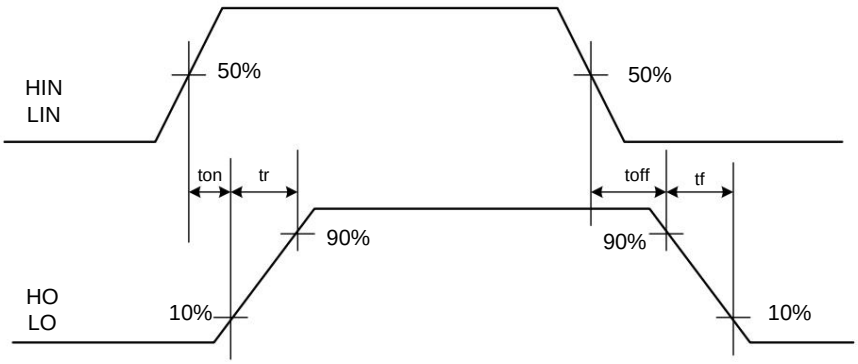
parameter	symbol	Test Conditions	Min.	Typ.	Max.	Unit
Output rising edge transmission time	ton		-	300	450	ns
Output falling edge transmission time	toff		-	100	160	ns
Output rise time	tr	CL=1000pF	-	12	25	ns
Output fall time	tf	CL=1000pF	-	12	25	ns
High and low side delay matching	MT		-	-	30	ns
Dead time	DT		100	200	300	ns

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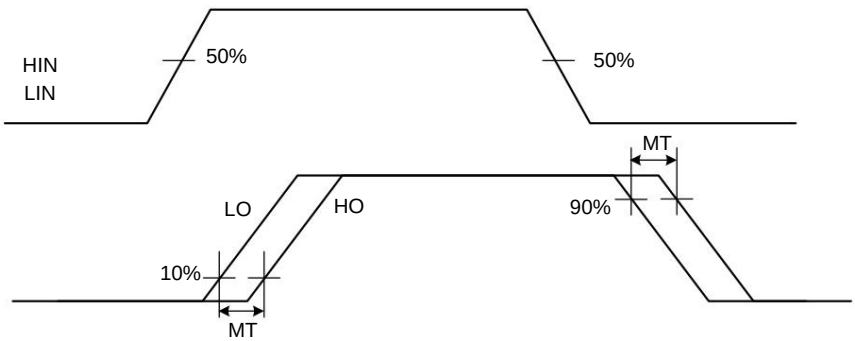
4 Logic function timing diagram



5 Transmission time test standards



6 Transmission Time Matching Test Standards

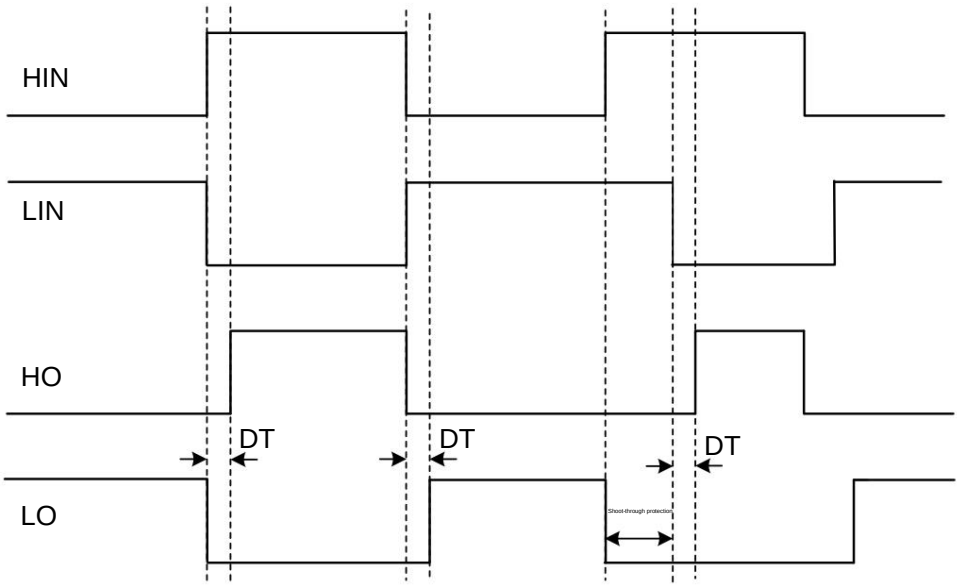


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7. Through prevention function

The chip is designed with a protection circuit specifically designed to prevent the power tube from being directly connected, which can effectively prevent the high-side and low-side input signals from being interfered with.

The following figure shows how the shoot-through prevention circuit protects the power tube.

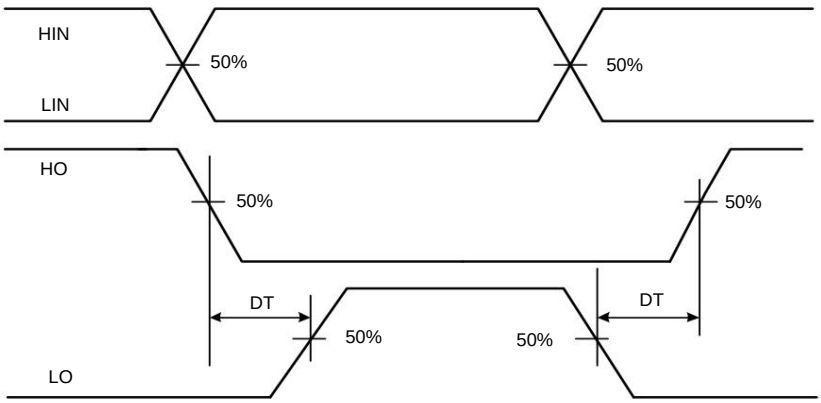


8 Dead zone function

A fixed dead time protection circuit is set inside the chip. During the dead time, both the high-side and low-side outputs are set to low level.

The dead time set must ensure that after one power tube is turned off, another power tube is turned on to effectively prevent the generation of up and down power

The following figure describes the timing relationship between the dead time, input signal and driver output signal.



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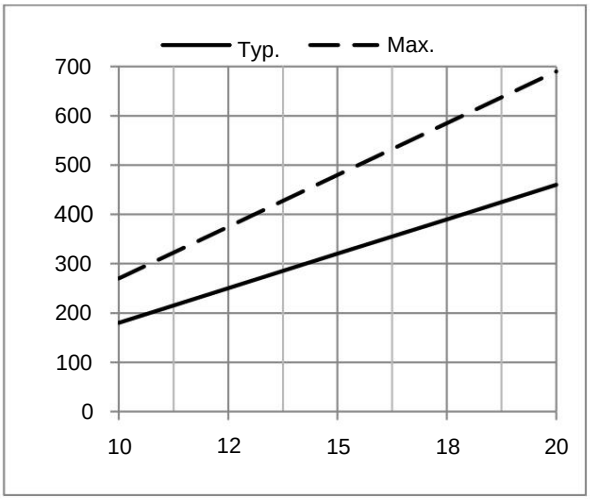


Figure 1A VCC supply current vs VCC supply voltage

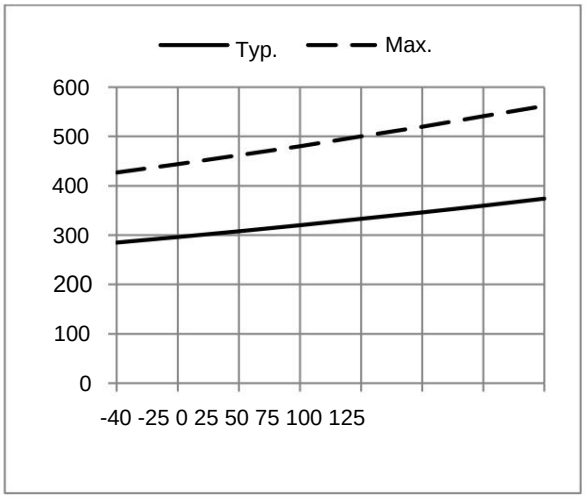


Figure 1B VCC supply current vs temperature

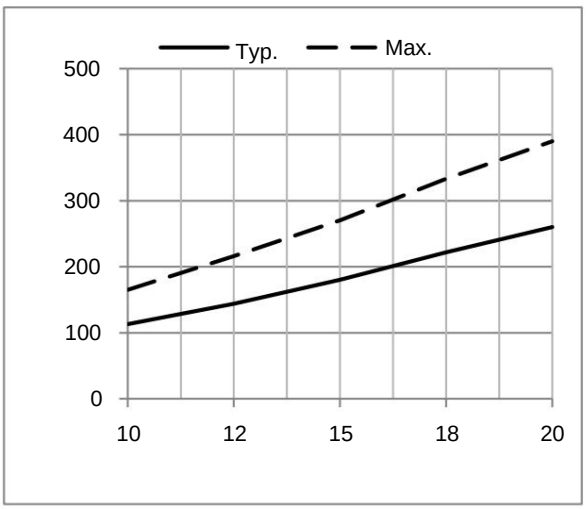


Figure 2A VBS supply current vs VBS supply voltage

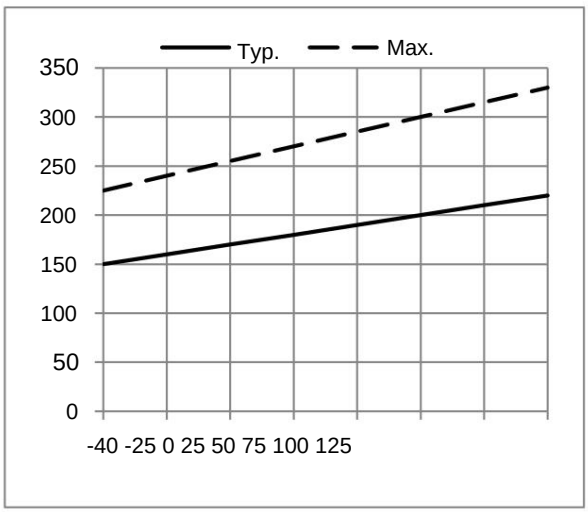


Figure 2B VBS supply current vs temperature

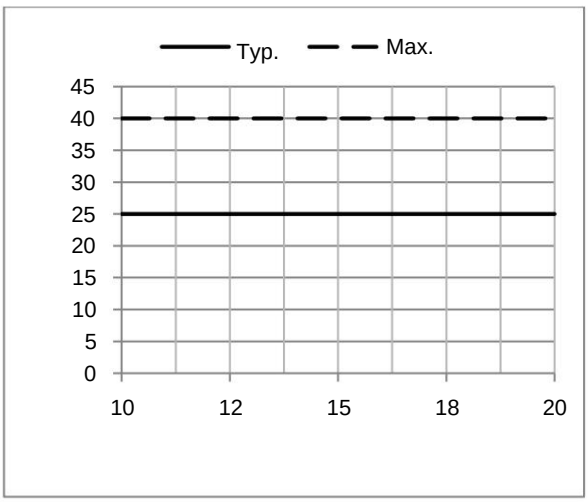


Figure 3A High-level input bias current vs supply voltage

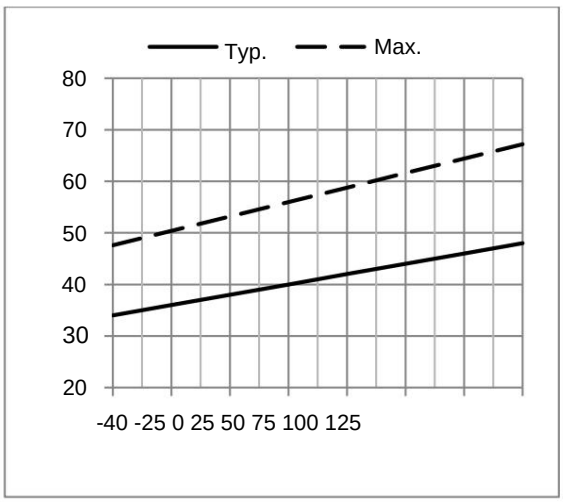


Figure 3B High-level input bias current vs. temperature

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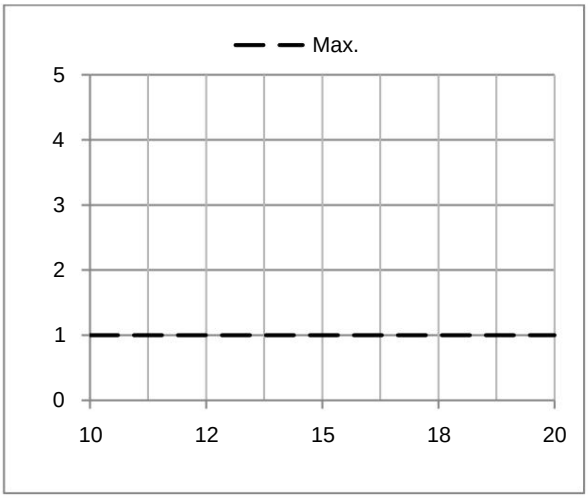


Figure 4A Low-level input bias current vs VCC supply voltage

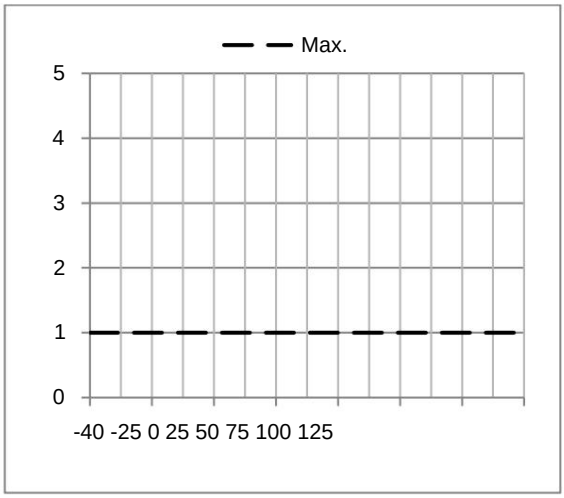


Figure 4B Low-level input bias current vs. temperature

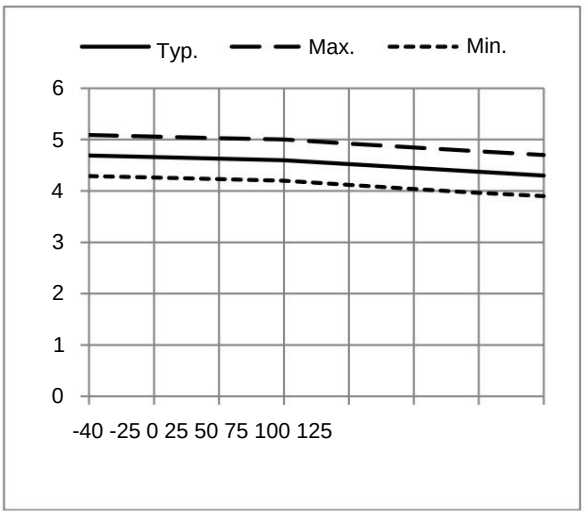


Figure 5A VCC undervoltage trip voltage vs temperature

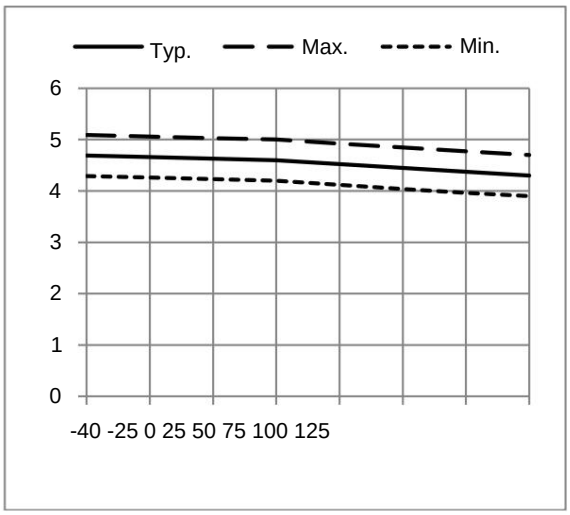


Figure 5B VCC undervoltage reset voltage vs temperature

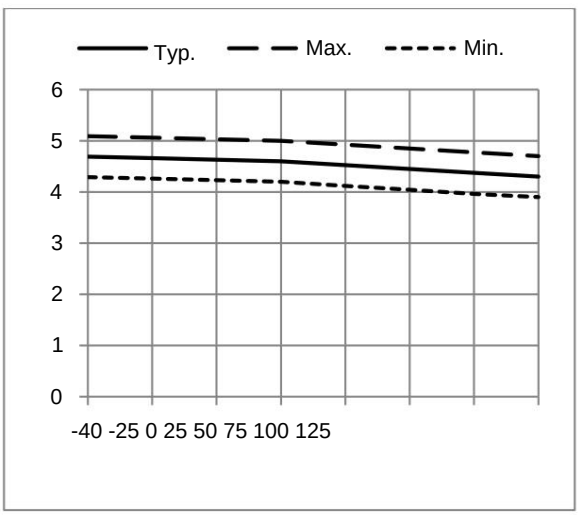


Figure 6A VBS undervoltage trip voltage vs temperature

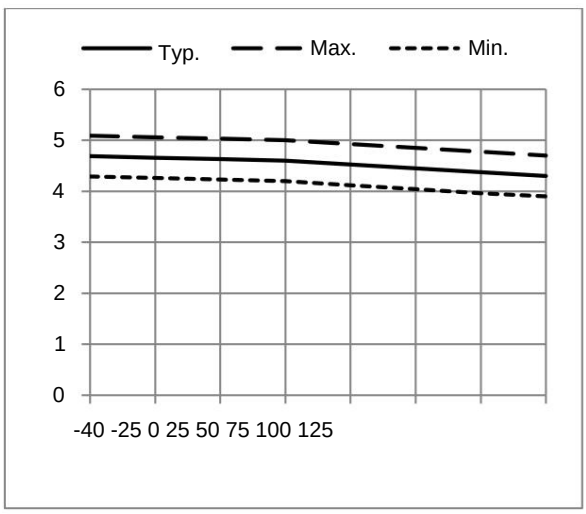


Figure 6B VBS undervoltage reset voltage vs temperature

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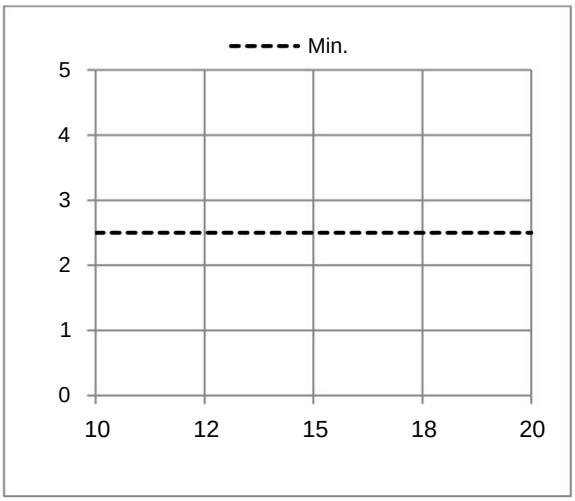


Figure 7A High-level input threshold voltage vs VCC supply voltage

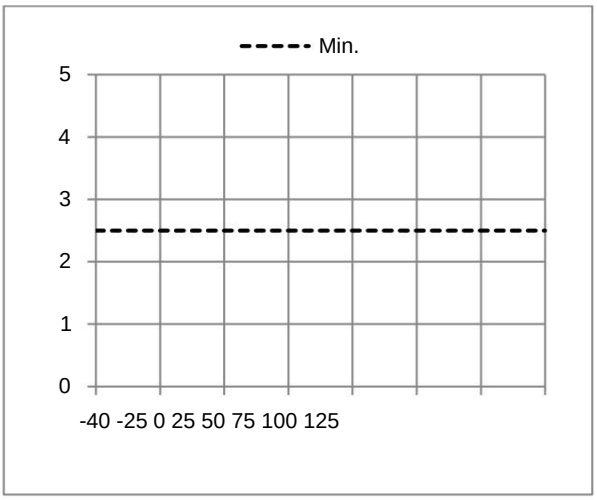


Figure 7B High level input threshold voltage vs temperature

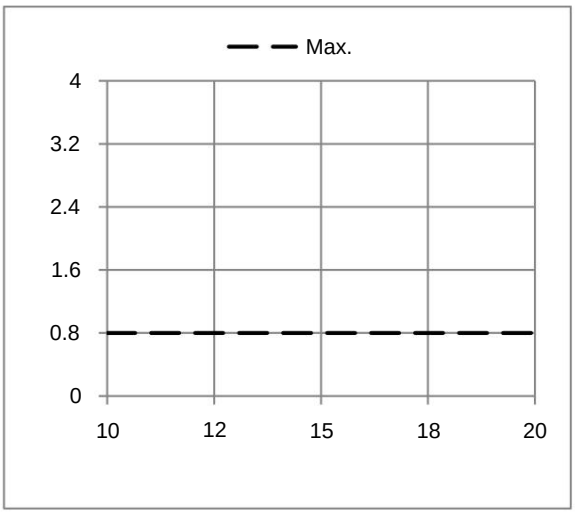


Figure 8A Low-level input threshold voltage vs. VCC supply voltage

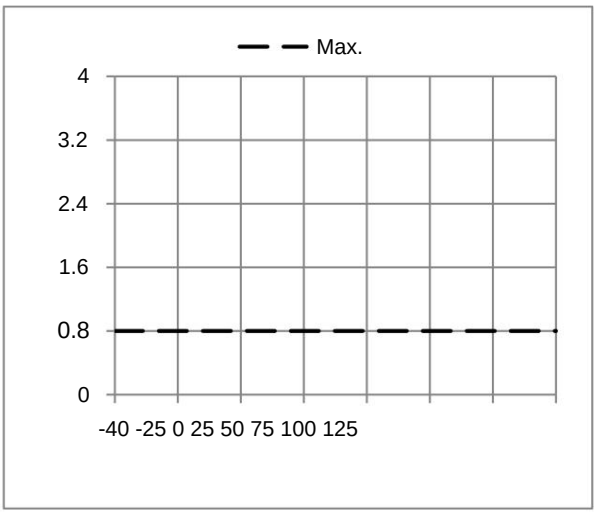


Figure 8B Low level input threshold voltage vs temperature

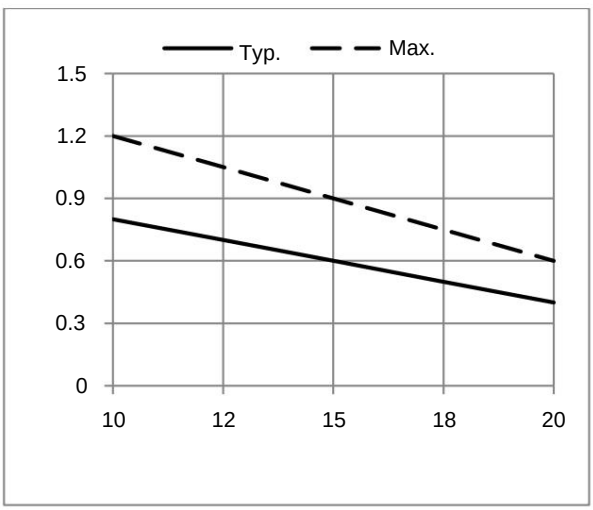


Figure 9A High-level output voltage vs. supply voltage

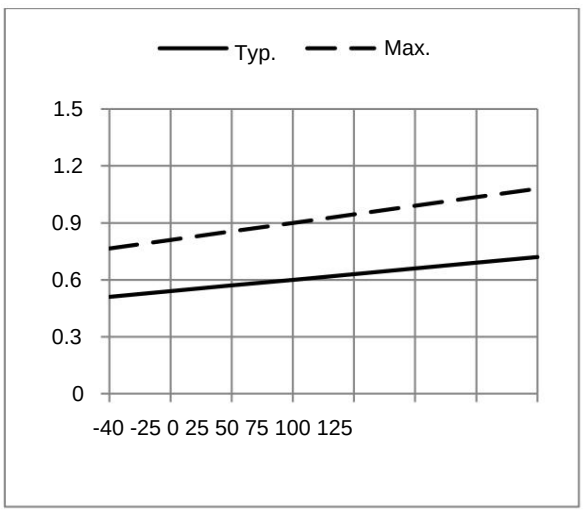


Figure 9B High level output voltage vs temperature

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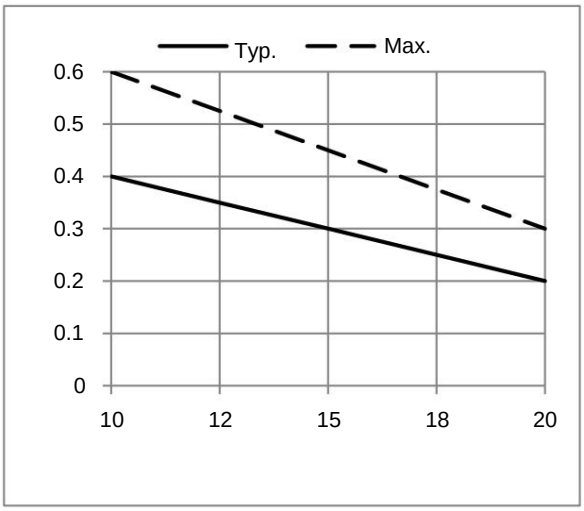


Figure 10A Low-level output voltage vs. supply voltage

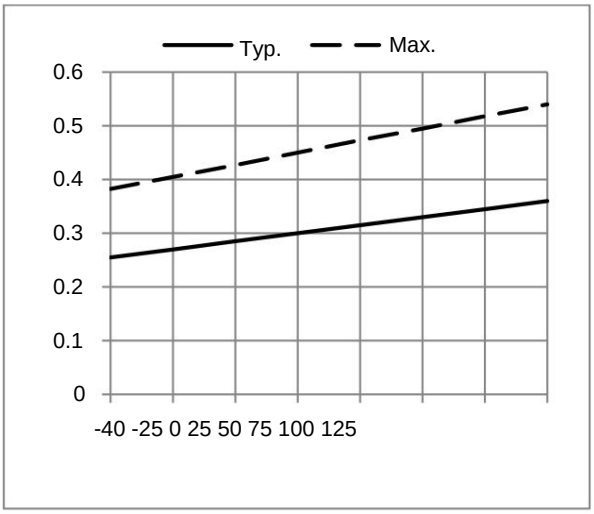


Figure 10B Low level output voltage vs temperature

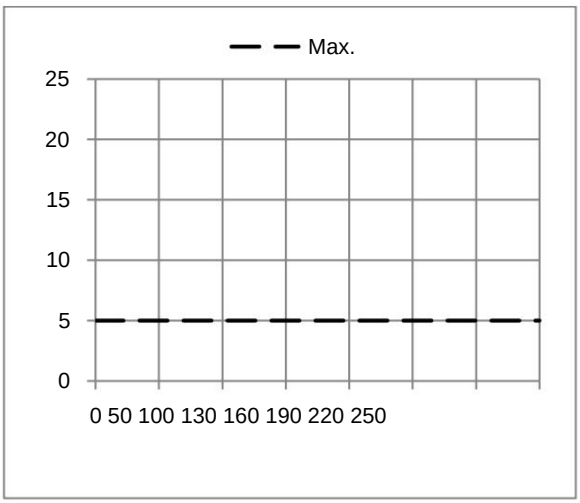


Figure 11A Floating supply leakage current vs supply voltage

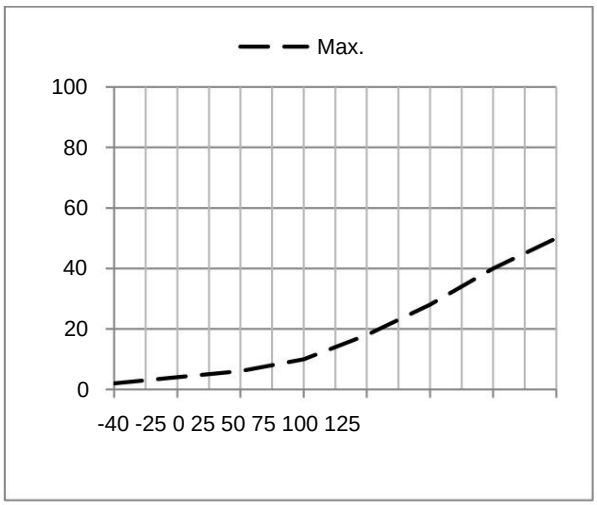
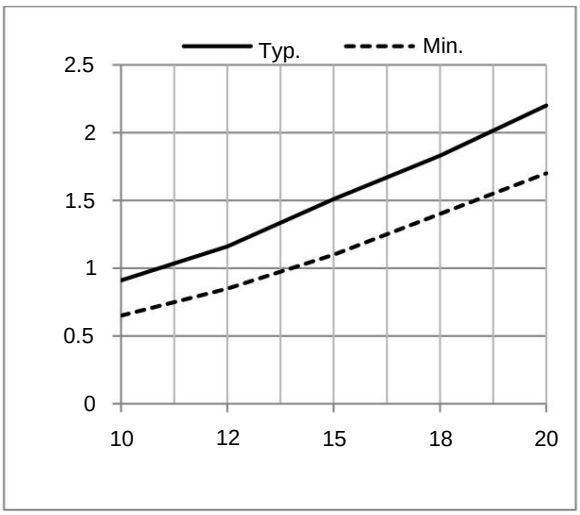


Figure 11B Floating power supply leakage current vs temperature



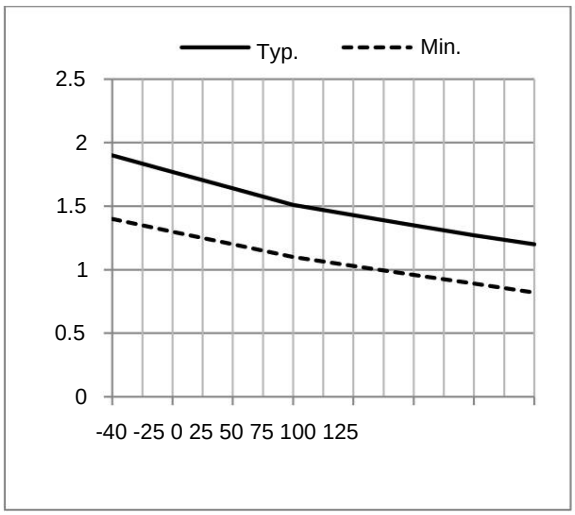
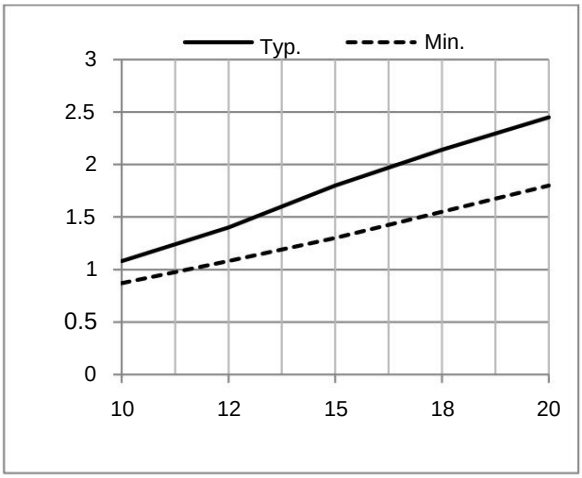


Figure 12A High level output short circuit pulse current vs power supply voltage Figure 12B High level output short circuit pulse current vs temperature

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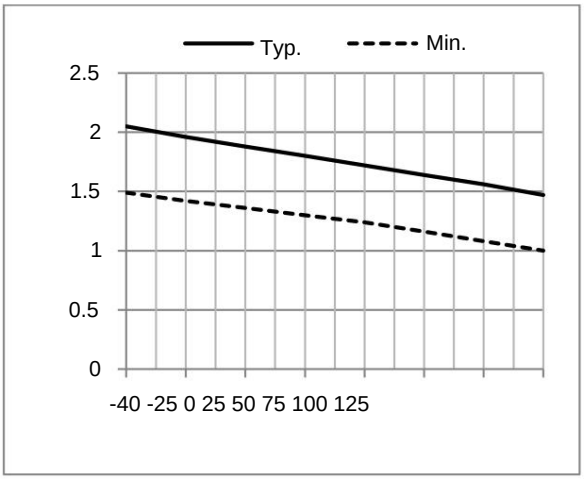
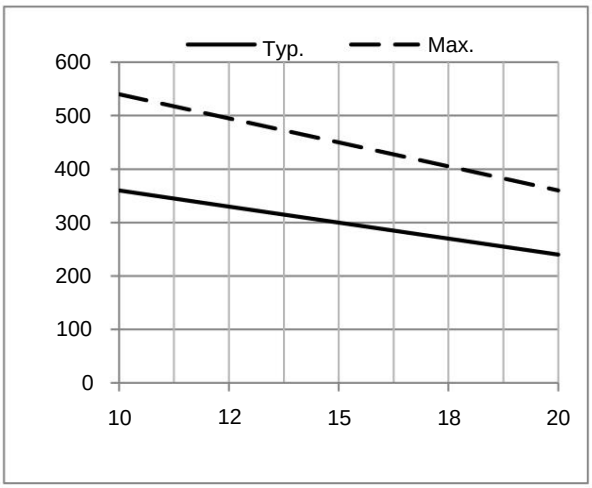


Figure 13A Low-level output short-circuit pulse current vs. power supply voltage Figure 13B Low-level output short-circuit pulse current vs. temperature



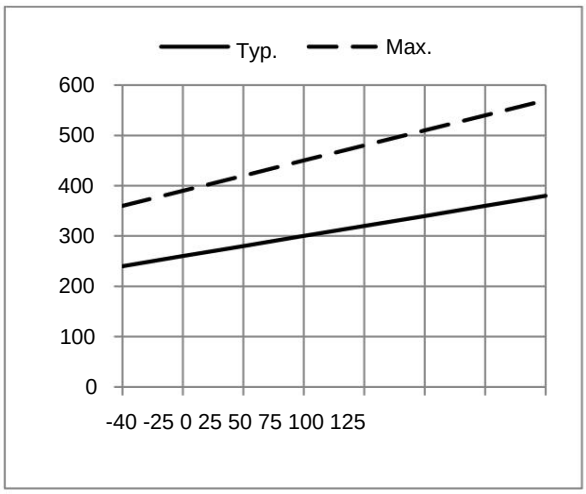
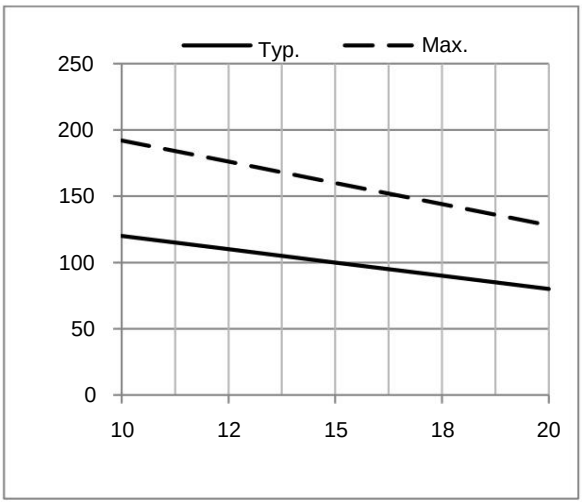


Figure 14A Output rising edge propagation time vs supply voltage

Figure 14B Output rising edge transmission time vs temperature



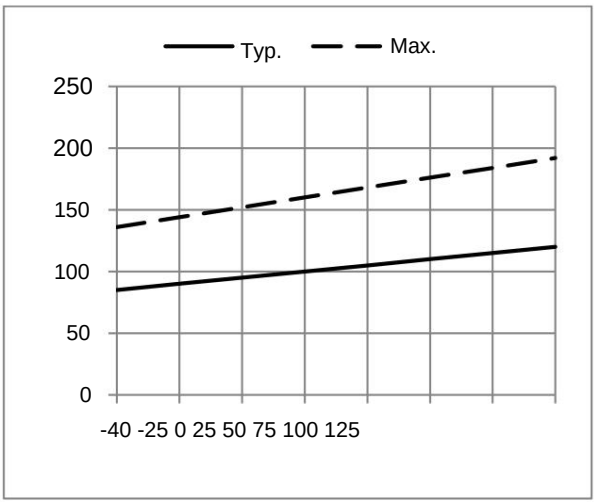


Figure 15A Output falling edge propagation time vs supply voltage

Figure 15B Output Falling Edge Transmission Time vs Temperature

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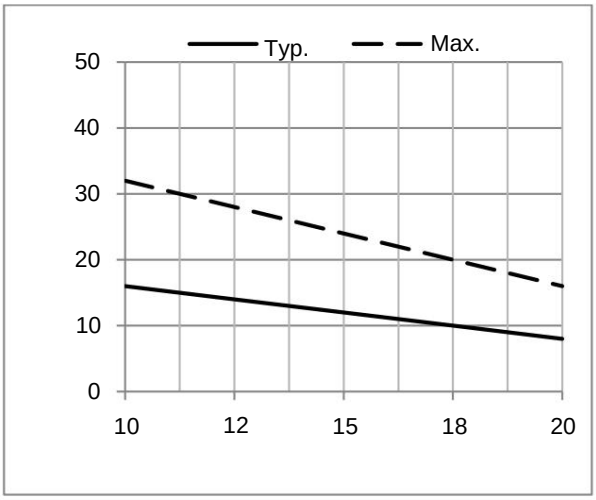


Figure 16A Rise time vs supply voltage

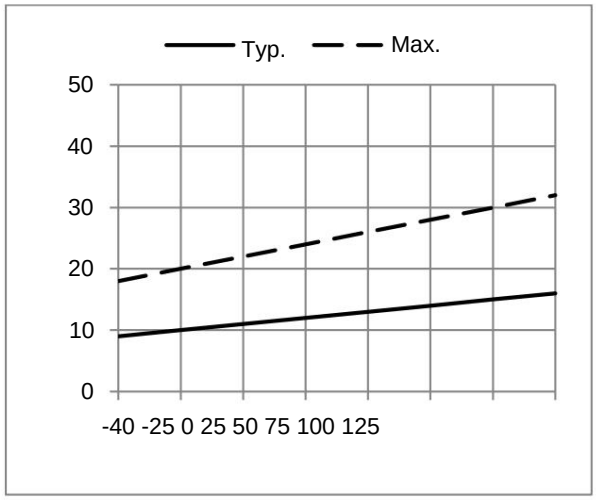


Figure 16B Rise time vs temperature

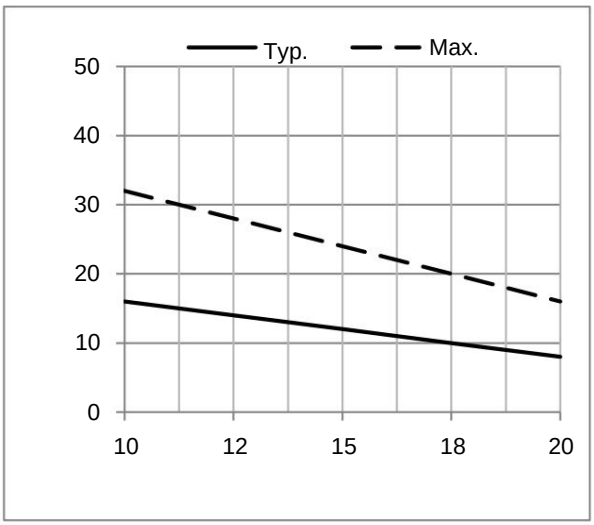


Figure 17A Fall Time vs Supply Voltage

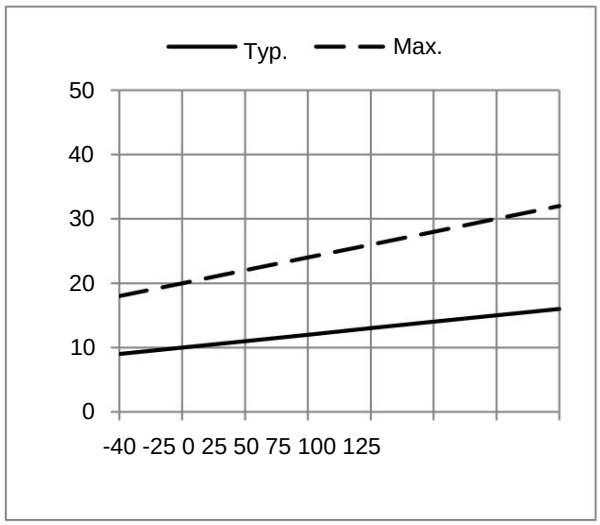


Figure 17B Fall time vs temperature

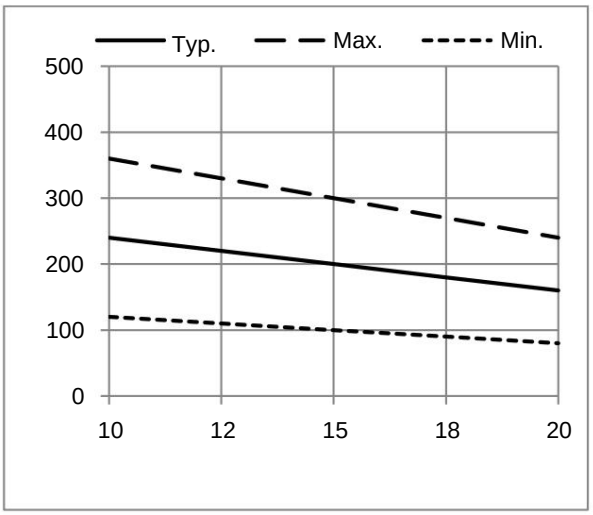


Figure 18A Dead time vs supply voltage

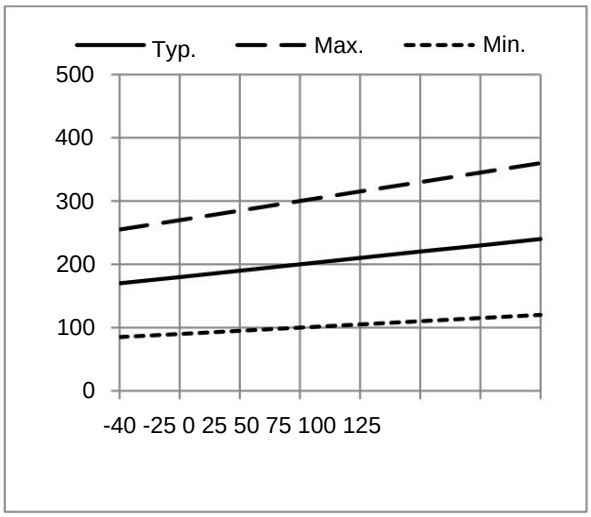


Figure 18B Dead time vs temperature

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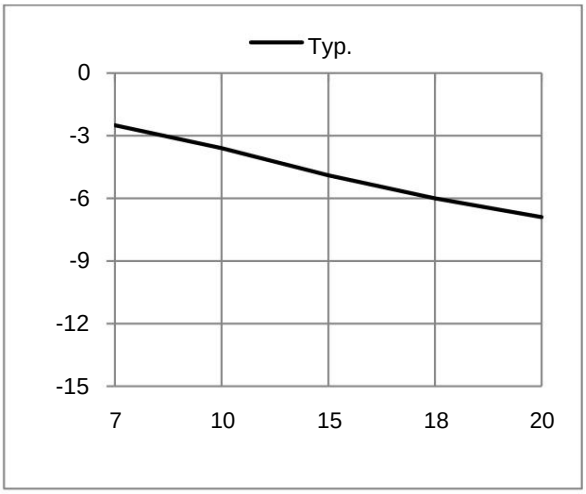


Figure 19A VS static negative pressure vs supply voltage

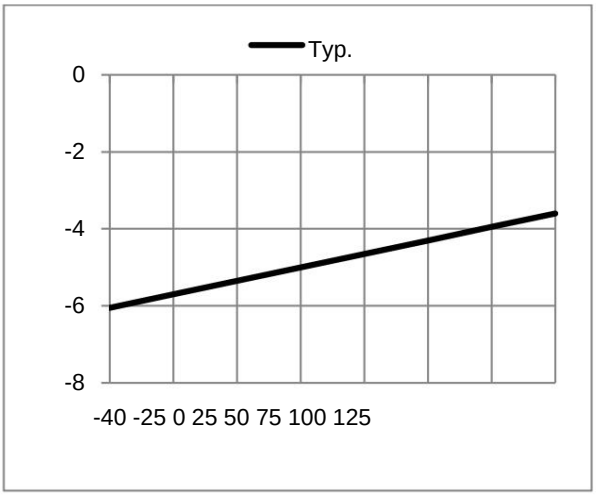


Figure 19B VS static negative pressure vs temperature

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9 Modification Record

Version	Main changes	Effective Date	Modified by
V0.1	First version	2015/05/26	Cheng Chunyun
V0.2	Added parameters: VS static negative voltage typical value is -4.5V, maximum value is -3V	2015/06/08	Cheng Chunyun
V0.3	adds dead time, shoot-through prevention, VS dynamic negative pressure and function description	2015/07/29	Cheng Chunyun
V0.4	Added QFN24 package, undervoltage adjustment, parameter reference voltage adjusted from 12V to 15V	2015/12/09	Xie Zhengkai
V0.5	Added QFN24 package pin diagram	2015/12/21	Cheng Chunyun
V0.6	VCC/VB undervoltage typical value is adjusted	2016/02/20	Xie Zhengkai
V0.7	Modify the package pin	2016/03/18	Xie Zhengkai
V0.8	Added suffix to product model, FD6288 changed to FD6288T&Q V0.9 Removed	2016/12/18	Xie Zhengkai
top screen printing, added packaging information		2018/01/05	Xie Zhengkai
V1.0	Dead time description and static negative pressure modification	2020/05/04	Xie Zhengkai
V1.1	Added R2 and related instructions to the typical application circuit	2023/04/24	Ye Lei
V1.2	"Switching time test standard" was changed to "Transmission time test standard"	2023/04/26	Ye Lei/ Deng Shuzhi
V1.3	Added electrical parameter curve	2023/05/23	Ye Lei/ Deng Shuzhi
V1.4	Added logic function timing diagram	2023/05/26	Ye Lei/ Deng Shuzhi
V1.5	Document format standardization	2023/05/29	Deng Shuzhi
V1.6	1. Updated 1.6 functional block diagram; 2. Modify 2 packaging information diagram; 3.3.2 Recommended Operating Conditions Note [3] Delete "VS1,2,3 is (COM-4V) to (COM-VBS), the HO logic state is maintained"; 4. The entire text was changed from FD6288T&Q to FD6288; 5. The symbol for microampere is unified as ÿA; 6. Unify the document serial number 12 to 28.	2023/08/31	Zhu Binghua

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